

2.4 Partial Fractions Worksheet

1. Compute $\int \frac{3x}{x^2 - 2x - 8} dx$

2. Compute $\int_2^3 \frac{x-4}{x^3-x^2} dx$

3. Compute $\int \frac{2x^2 + 5x + 12}{x^3 + 2x^2 + x + 2} dx$

4. Compute $\int_{-5}^{-1} \frac{x+2}{x^3 - 2x^2 + x} dx$

5. Compute $\int \frac{x^3 + 4x^2 + 2x - 1}{x^2 + x - 2} dx$

6. Compute $\int \frac{5}{(x^2 + 4)(x - 1)} dx$

7. Find $\int \frac{x^4 + x^3 + 2x^2 + 2x + 1}{x^3 + x^2} dx$.